Python type casting

In [1]:
```
110.34
```
Out[1]: 110.34

In [2]:
```
int(110.34)                    #we convert float ti int
```
Out[2]: 110

In [3]:
```
int(True)
```
Out[3]: 1

In [4]:
```
int(1+2j)
```
```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[4], line 1
----> 1 int(1+2j)

TypeError: int() argument must be a string, a bytes-like object or a real number,
not 'complex'
```

In [5]:
```
int('10')
```
Out[5]: 10

In [6]:
```
int('ten')
```
```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[6], line 1
----> 1 int('ten')

ValueError: invalid literal for int() with base 10: 'ten'
```

Other Data type to Float

In [7]:
```
float(110)
```
Out[7]: 110.0

In [8]:
```
float(True)
```
Out[8]: 1.0

In [9]:
```
float(False)
```
Out[9]: 0.0

In [10]:
```
float(1+2j)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[10], line 1
----> 1 float(1+2j)

TypeError: float() argument must be a string or a real number, not 'complex'
```

In [11]:
```python
float('10')                    #Number string
```
Out[11]: 10.0

In [12]:
```python
float('ten')                   #text string
```
```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[12], line 1
----> 1 float('ten')

ValueError: could not convert string to float: 'ten'
```

Other Data type to Boolean

In [13]:
```python
bool(2)
```
Out[13]: True

In [14]:
```python
bool(0)
```
Out[14]: False

In [15]:
```python
bool(2.3)
```
Out[15]: True

In [16]:
```python
bool(1+2j)
```
Out[16]: True

In [17]:
```python
bool('10')
```
Out[17]: True

In [18]:
```python
bool('ten')
```
Out[18]: True

In [19]:
```python
bool()
```
Out[19]: False

In [20]:
```python
bool(2-)
```
```
  Cell In[20], line 1
    bool(2-)
           ^
SyntaxError: invalid syntax
```

All Other Data Type to Complex

In [21]:
```
complex(2)
```
Out[21]:
```
(2+0j)
```

In [22]:
```
complex(3,4)
```
Out[22]:
```
(3+4j)
```

In [23]:
```
complex(3,4,5)
```
```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[23], line 1
----> 1 complex(3,4,5)

TypeError: complex() takes at most 2 arguments (3 given)
```

In [24]:
```
complex(2.3,4.5)
```
Out[24]:
```
(2.3+4.5j)
```

In [25]:
```
complex(True)
```
Out[25]:
```
(1+0j)
```

In [26]:
```
complex(False)
```
Out[26]:
```
0j
```

In [27]:
```
complex('10')
```
Out[27]:
```
(10+0j)
```

In [28]:
```
complex('ten')
```
```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[28], line 1
----> 1 complex('ten')

ValueError: complex() arg is a malformed string
```

In [29]:
```
complex('10','20')
```
```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[29], line 1
----> 1 complex('10','20')

TypeError: complex() argument 'real' must be a real number, not str
```

In [30]:
```
str(10)
```
Out[30]:
```
'10'
```

In [31]:
```
str(10.4)
```
Out[31]:
```
'10.4'
```
In [32]:
```
str(1+2j)
```
Out[32]:
```
'(1+2j)'
```
In [34]:
```
str(True)
```
Out[34]:
```
'True'
```
In [ ]:
```

```